

PRD: HRIS Role Mapping

1. Overview

When organizations connect their HRIS (Human Resource Information System) to 5mins via Merge.dev, synced employees arrive without a job role. Today, every one of those employees must manually pick a role during onboarding — a friction point that delays access to relevant learning paths and scales painfully for large customers syncing hundreds or thousands of users.

This feature introduces **automatic HRIS-to-5mins role mapping**: during the existing Dry Run phase, the system fetches HRIS job titles, attempts to match them to 5mins roles, and presents the results in a reviewable mapping table. Admins refine the mappings, and when the integration switches to Full or Partial sync, users are invited with their roles pre-assigned — skipping the role-selection step entirely during onboarding. On subsequent syncs, title changes are detected and roles updated accordingly.

The core **job to be done** is straightforward: large-org admins need their HRIS investment to flow through to learning-platform readiness without per-user manual intervention.

2. Jobs To Be Done

Main Job

Ensure every HRIS-synced employee lands in the correct 5mins learning path automatically, so that neither the admin nor the employee wastes time on manual role selection.

Job Map

Stage	What Happens Today	Pain / Workaround	What This Feature Does
Define	Admin decides which employees need which learning paths	No way to express intent at scale — role assignment is per-user	Mapping table lets admin define intent once per job title
Locate	Admin connects HRIS via Merge.dev Dry Run	Dry Run fetches employees but ignores job titles	Dry Run now fetches distinct titles + counts
Prepare	N/A — no preparation step exists	Admin has no preview of what roles users will get	Auto-matching computes suggested mappings; admin reviews
Confirm	Admin switches Dry Run → Full/Partial	No role information is carried into the sync	Mappings are stored and applied when sync activates

Execute	Invites sent; users onboard and manually pick roles	Every user hits the role-picker — slow, error-prone, inconsistent	Users with mapped roles skip role selection entirely
Monitor	Admin has no visibility into role assignment status	No way to know if roles are correct post-sync	Mapping table shows statuses (Mapped / Unmapped / Broken); banner alerts for new titles
Resolve	Admin manually reassigns roles one-by-one	Unscalable for large orgs	Admin edits mappings; next sync propagates changes

Related Jobs

- **IT Admin:** "Keep our HRIS as the single source of truth for employee data across all SaaS tools."
- **L&D Manager:** "Get employees into relevant learning content on day one, not day five."
- **End User (Employee):** "Start learning immediately without bureaucratic setup steps."

Emotional & Social Dimensions

- **Admin** wants to feel **in control and confident** — they need to trust that the system won't silently assign wrong roles to hundreds of people. The Dry Run → review → activate flow gives them a safety net.
- **Admin** wants to be perceived as **efficient and competent** by their organization — automating role assignment at scale demonstrates operational sophistication.
- **Employee** wants to feel **expected and welcomed** — landing directly in a personalized learning path (rather than being asked "what do you do?") signals that the organization prepared for their arrival.

3. Goals

1. **Zero-touch role assignment for mapped titles** — employees whose HRIS title has a mapping should never see the role-selection step during onboarding.
2. **High auto-match rate on first run** — case-insensitive exact matching against tenant roles (priority) then public roles should resolve the majority of titles without admin intervention.
3. **Clear admin triage workflow** — admins should be able to scan, filter, and resolve all unmapped or broken titles in a single session, with confidence in what each action does.
4. **Safe re-sync behavior** — title changes propagate correctly; unmapped titles never silently remove an existing role; broken mappings are surfaced, not hidden.
5. **Prototype scope: ~20 job titles** — the initial implementation should be designed to handle dozens to hundreds of titles, but the prototype will target roughly 20 for validation.

4. Job Stories

Admin — Initial Setup

- **When** I connect my HRIS in Dry Run mode, **I want** the system to automatically match HRIS job titles to 5mins roles, **so I can** review suggested mappings instead of building them from scratch.
- **When** I see the mapping table after Dry Run, **I want** to filter to only unmapped titles, **so I can** focus my time on the titles that need my attention.
- **When** I see an auto-matched title that maps to the wrong 5mins role, **I want** to change the mapping via an inline dropdown, **so I can** correct it without navigating away.

Admin — Ongoing Management

- **When** a re-sync discovers new HRIS job titles, **I want** to see a specific in-app banner (e.g., "3 new job titles discovered — 12 employees need role mapping"), **so I can** act on it promptly rather than discovering gaps later.
- **When** a 5mins role that was part of a mapping gets deleted, **I want** the mapping to show as "Broken" with a visual indicator, **so I can** reassign those titles before the next sync.
- **When** I remove a mapping, **I want** existing users to keep their current role, **so I can** clean up mappings without disrupting active learners.

Employee — Onboarding

- **When** my role has been pre-assigned from HRIS, **I want** to skip the role-selection step during onboarding, **so I can** start learning immediately.
- **When** my HRIS title has no mapping, **I want** to be prompted to pick a role during onboarding, **so I can** still get a personalized experience even if my title wasn't pre-mapped.

Admin — Conflict Resolution

- **When** a user's HRIS title changes and the new title has a mapping, **I want** the system to update their role automatically, **so I can** keep learning paths aligned with organizational changes.
- **When** a user's HRIS title changes to an unmapped title, **I want** the system to keep their existing role and flag the new title as unmapped, **so I can** avoid disrupting the user's learning while I review the new title.

5. Requirements

Functional Requirements

FR-1: Dry Run Enhancement — Title Fetching & Auto-Matching

- **FR-1.1:** During Dry Run, the system SHALL fetch all employees from the connected HRIS via Merge.dev and compute a list of distinct job titles with associated employee counts.
- **FR-1.2:** The system SHALL auto-match each distinct HRIS title to a 5mins role using case-insensitive exact string match, checking tenant-specific roles first, then public roles. (Tenant role wins in case of conflict per clarifying answer.)
- **FR-1.3:** The system SHALL store the resulting mappings (matched and unmatched) as the tenant's HRIS role mapping configuration.
- **FR-1.4:** Dry Run SHALL NOT create invites, users, or team assignments.

FR-2: Admin Mapping Table (Integration Settings Screen)

- **FR-2.1:** The mapping table SHALL display: HRIS Job Title, Employee Count, Mapped 5mins Role (with Edit/Remove actions), and Status (Mapped / Unmapped / Broken).
- **FR-2.2:** Status SHALL use color-coded badges: green for Mapped, amber/gray for Unmapped, red for Broken.
- **FR-2.3:** Admins SHALL be able to change any mapping at any time by selecting a different 5mins role from an inline dropdown (tenant and public roles available).
- **FR-2.4:** Admins SHALL be able to remove a mapping. Removing a mapping SHALL NOT change existing users' roles; future syncs for that title will not assign a role.
- **FR-2.5:** The table SHALL support a filter/tab to show only unmapped titles.
- **FR-2.6:** When a mapped 5mins role is deleted from the system, the mapping status SHALL change to "Broken."
- **FR-2.7:** The table SHALL support frozen headers for scrollability.

FR-3: Notifications for New Titles

- **FR-3.1:** When a re-sync discovers new HRIS job titles not present in the current mapping configuration, those titles SHALL be added as unmapped.
- **FR-3.2:** An in-app informational banner SHALL appear on the integration settings screen, specifying the count of new titles and affected employees (e.g., "3 new job titles discovered — 12 employees need role mapping").

FR-4: Sync Activation (Full / Partial)

- **FR-4.1:** When a SuperAdmin switches from Dry Run to Full or Partial sync, the initial sync SHALL apply mapped roles to corresponding users at invite time.
- **FR-4.2:** Users whose HRIS title is unmapped SHALL proceed through the standard onboarding role-selection flow.

FR-5: Re-Sync Role Update Logic

- **FR-5.1:** The system SHALL store `hrisJobTitle` on each user's profile.
- **FR-5.2:** On each sync, the system SHALL compare the incoming HRIS title against the stored `hrisJobTitle`.

- **FR-5.3:** If the title has changed AND the new title has a mapping, the system SHALL update the user's 5mins role and trigger the existing skill-update behavior.
- **FR-5.4:** If the title has changed AND the new title has NO mapping, the system SHALL keep the user's existing role and add the new title as unmapped in the mapping table.
- **FR-5.5:** If the title has NOT changed, no role update SHALL occur.

FR-6: Edge Cases

- **FR-6.1:** Empty HRIS title → no role assigned; user selects role during onboarding.
- **FR-6.2:** Inactive HRIS employee → skipped entirely (existing behavior).
- **FR-6.3:** Users SHALL NOT be able to change their HRIS-assigned role themselves; only admins can change it (per clarifying answer).

FR-7: Onboarding UX — Skip Role Selection (Deliverable 2)

- **FR-7.1:** When a user's role has been pre-assigned via HRIS mapping, the role-selection step in onboarding SHALL be skipped entirely.
- **FR-7.2:** Users with no mapped role (unmapped or empty title) SHALL see the standard role-selection step.

6. Research & Best Practices

Synthesis

The mapping table is the centrepiece of the admin experience, and research strongly supports the design decisions already outlined in the ticket.

Table design fundamentals. NN/g's research on data tables identifies four core user tasks — finding records, comparing data, viewing/editing records, and taking batch actions — all of which apply to the mapping table ([NN/g — Data Tables](#)). Frozen headers, human-readable status labels (not codes), and discoverable filters are called out as essential for tables with 20+ rows. Pencil & Paper's enterprise data table analysis recommends inline editing via dropdowns as the lowest-friction pattern for reassigning values like roles, and suggests sidebar panels when admins need to drill into detail (e.g., viewing which employees hold a given title) ([Pencil & Paper — Data Tables](#)).

Filtering patterns. The ticket's requirement to "filter to show only unmatched titles" aligns precisely with Pencil & Paper's recommendation for pre-filtered tabs when filter sets grow large, especially when connected to HR databases ([Pencil & Paper — Filtering](#)). Visible filter tags and search-within-filter are also recommended as the mapping table scales.

SaaS user management conventions. Perpetual's guide to SaaS user management advocates color-coded pill badges for status columns and context-aware actions — e.g., "Map" for unmapped, "Edit" for mapped, "Fix" for broken — which directly maps to our three statuses ([Perpetual — SaaS User Management](#)). Success notifications should confirm exact changes ("3 titles mapped successfully"), and empty states after filtering should say "No unmapped titles found" with a clear-filter action.

HRIS integration realities. Apideck's guide to HRIS integrations warns that field naming varies wildly across providers and recommends creating a data dictionary of field mappings and cleaning data before syncing ([Apideck — HRIS Guide](#)). This validates the decision to start with case-insensitive exact match and defer fuzzy matching. Merge.dev's syncing best practices recommend combining webhooks with polling and using `modified_after` timestamps for incremental syncs ([Merge.dev — Syncing Best Practices](#)). Crucially, each Employee can have multiple Employment objects — when a title changes, a new Employment is created with an `effective_date`, so the re-sync logic should resolve to the most recent employment's title. Merge.dev's HRIS API overview confirms that the `job_title` field is passed through as-is when no enum mapping exists, validating the approach of displaying raw titles for admin review ([Merge.dev — HRIS API](#)).

Dry Run pattern. Google Cloud's dry run mode for VPC Service Controls provides a strong precedent: configurations are created and logged but not enforced until activated ([Google Cloud — Dry Run](#)). This validates the ticket's Dry Run approach of fetching and computing mappings without creating invites or assignments.

Inline editing and undo. Eleken's table design guide recommends inline editing with dropdowns for role/status assignment and undo/redo for inline edits, with non-blocking error highlights ([Eleken — Table Design](#)). The SaaS UI Workflow Patterns gist notes that admin views combining navigation, data tables, filters, and action buttons form a foundational SaaS pattern, and that split-pane/master-detail views support before/after comparison useful for dry run previews ([SaaS UI Patterns Gist](#)).

UX References

App	Flow / Screen	URL	Pattern Description	Relevance
Attio	Importing Data (Web)	mobbin.com/flows/81dfb7b0...	13-screen CSV import: file selection → attribute mapping (column-to-field matching) → review/preview → confirmation → processing. Validation at each step.	Directly analogous to HRIS title → role mapping. The attribute mapping step mirrors matching titles to roles; the review step before committing mirrors Dry Run → Full transition.

Luma	Importing People (Web)	mobbin.com/explore/flows/7bcf9bd6...	12-screen people import: CSV upload → data preview with tagging → updated contact list. Bulk data review before commit.	Relevant to admin reviewing HRIS-synced titles with counts. Tagging step parallels assigning roles to title groups.
Slite	Importing a Doc (Web)	mobbin.com/explore/flows/255d2fa9...	9-screen doc import: source selection → channel configuration → progress indicators → results view. Clear step progression.	Demonstrates best practice for multi-step import with progress indicators — applicable to Dry Run fetching state and sync progress.

Built for Mars Lessons

Title	Type	URL	Lesson	Relevance
Slack Hides Complexity to Create Effortless Onboarding	Case Study	builtformars.com/case-studies/slack	Incrementally expose complexity rather than showing everything at once. Hiding complexity creates a smooth first run but can lead to cumulative noise.	Supports the phased Dry Run → review → Full flow. The "filter to unmapped only" pattern correctly stages admin attention rather than overwhelming with all titles at once.
How to Stop People from Skipping Your Onboarding (YNAB)	Case Study	builtformars.com/case-studies/ynab	Users have action bias — they prefer progress-feeling CTAs over pausing to learn. When role is pre-assigned, skipping the role step removes friction users would have rushed through anyway.	Validates "skip role selection when pre-set" for Deliverable 2. HRIS-assigned users should go straight to learning content.
Linear's "undo" Warnings	UX Bite	builtformars.com/ux-bites/linears-undo-warnings	Only show undo confirmation warnings when the action is contextually risky (>10	When admin removes a mapping affecting many employees, show a confirmation.

			min old or different page), not for every action. Prevents modal fatigue.	For routine single-title edits, skip the confirmation to avoid alert fatigue.
Framing the Value of Notifications (Citymapper)	UX Bite	builtformars.com/ux-bites/framing-the-value-of-notifications	Frame notification value with specific examples, not generic copy.	In-app banners for new unmatched titles should be specific: "3 new job titles discovered — 12 employees need role mapping" rather than generic "New unmatched titles."
Timed Custom Statuses (Discord)	UX Bite	builtformars.com/ux-bites/timed-custom-statuses	Color-coded, clearly differentiated badges allow at-a-glance scanning without reading labels.	Directly applicable to Mapped (green) / Unmapped (amber) / Broken (red) status badges in the mapping table.

7. Plan of Action

Phase 1: Backend — Mapping Engine & Data Model

- ☐ Design data model for HRIS role mappings: `hris_role_mapping` table (`tenant_id`, `hris_job_title`, `fivemins_role_id`, `status`, `employee_count`, `created_at`, `updated_at`)
- ☐ Add `hrisJobTitle` field to user profile schema
- ☐ Extend Dry Run logic to fetch distinct job titles and employee counts from Merge.dev Employment objects (resolving to most recent `effective_date` per employee)
- ☐ Implement auto-match algorithm: case-insensitive exact match against tenant roles first, then public roles
- ☐ Build CRUD API endpoints for mapping management (list, update, remove mappings)
- ☐ Implement re-sync comparison logic: detect title changes, resolve mappings, update roles or flag unmapped
- ☐ Handle edge cases: empty titles, inactive employees, deleted roles (mark mapping as Broken)
- ☐ Write unit and integration tests for all mapping engine logic

Phase 2: Admin UI — Mapping Table (Deliverable 1)

- ☐ Request Figma references from design team for mapping table components (per clarifying answer: reuse existing design system)
- ☐ Build mapping table component on integration settings screen: HRIS Job Title, Employee Count, 5mins Role (inline dropdown), Status (color-coded badge)
- ☐ Implement inline role dropdown populated with tenant + public roles
- ☐ Implement "Unmapped Only" filter tab/toggle
- ☐ Implement frozen table headers for scroll behavior
- ☐ Build in-app informational banner for new unmatched titles (specific copy with title count + employee count)
- ☐ Implement empty states: "No unmapped titles found" with clear-filter action; "No mappings yet — run Dry Run to discover titles"
- ☐ Add confirmation dialog for high-impact mapping removals (e.g., affecting many employees); skip for routine edits
- ☐ Build success toast notifications for mapping actions (e.g., "3 titles mapped successfully")
- ☐ Test admin flows end-to-end: Dry Run → review mappings → edit/remove → filter

Phase 3: Onboarding UX — Skip Role Selection (Deliverable 2)

- ☐ Modify onboarding flow to check for HRIS-assigned role on user profile
- ☐ If role is pre-assigned, skip the role-selection step entirely
- ☐ If role is not pre-assigned (unmapped/empty title), show standard role-selection step
- ☐ Ensure users cannot change their HRIS-assigned role from their profile (admin-only control)
- ☐ End-to-end validation: Dry Run → mapping review → Full sync → new user onboarding without role step

Phase 4: Polish & Hardening

- ☐ Test with ~20 job titles prototype dataset (per clarifying answer)
- ☐ Verify re-sync behavior: title changes with mapped roles, title changes with unmapped roles, no title changes
- ☐ Verify Broken status detection when a mapped 5mins role is deleted
- ☐ Verify banner notification accuracy after re-syncs with new titles
- ☐ Cross-browser and responsive testing for mapping table
- ☐ Performance check: ensure mapping table renders smoothly with hundreds of rows (future-proofing)
- ☐ Accessibility review: keyboard navigation for table, screen reader support for status badges

8. Risks & Mitigations

Risk	Impact	Mitigation
Low auto-match rate — HRIS titles are freeform and rarely match 5mins role names exactly (e.g., "Sr. SWE" vs "Software Engineer")	Admins face a mostly-unmapped table, reducing perceived value of the feature	Clearly communicate that exact match is V1; design the UI so manual mapping is fast (inline dropdowns, bulk review); track match rates to build the case for fuzzy matching in a future phase
HRIS title inconsistency across providers — different providers structure Employment objects differently; some may not populate <code>job_title</code> at all	Empty or missing titles create unmapped users, increasing admin workload	Handle empty titles gracefully (FR-6.1); log and surface providers with low title coverage; leverage Merge.dev's unified model to normalize where possible (Apideck — HRIS Guide)
Multiple Employment objects per employee — Merge.dev creates new Employment records for title changes; resolving to the "current" title requires correct <code>effective_date</code> logic	Wrong title resolved → wrong role assigned	Use the most recent Employment <code>effective_date</code> per Merge.dev best practices (Merge.dev — Syncing Best Practices); add logging for title resolution decisions
Stale Broken mappings go unnoticed — a 5mins role is deleted but the admin doesn't visit the integration settings page to see the Broken status	Users on subsequent syncs don't get roles assigned; no one notices	In-app banner for Broken mappings (same pattern as new-title banners); consider extending to email digest in a future phase
Admin confusion about mapping removal impact — admin removes a mapping thinking it will reset existing users' roles	Mismatched expectations; support tickets	Confirmation dialog for removals should explicitly state: "Existing users will keep their current role. Future syncs for this title will not assign a role."
Race condition between manual role edit and HRIS sync — admin edits a mapping while a sync is in progress	Sync could overwrite the just-edited mapping or apply stale data	Apply mapping changes atomically; sync reads mappings at start of run and applies consistently; document that in-progress syncs use the mapping state at sync start
Users locked out of role changes — HRIS-assigned users cannot change their role (FR-6.3), which may frustrate users whose HRIS title is wrong	User stuck in wrong learning path; support burden	Provide clear messaging to the user that their role is managed by their organization; direct them to their admin; ensure admins can quickly update individual mappings